Deloitte OA Workbook — Analyst (T&T; / Data & Analytics)

Comprehensive Notes, Real-style Practice Questions, SQL & Coding Exercises

Prepared for: Ramesh Chandra Soren

What's inside: Pattern overview, formula cheatsheets, 4 full mocks (25Q each), 60+ extra practice questions, 10 coding problems with solutions, 20 SQL challenges with solutions, study resources.

1. Deloitte OA — Pattern & What to Expect

Typical Deloitte online assessment for analyst roles usually combines: (1) Aptitude / Quantitative reasoning (number operations, data interpretation), (2) Logical reasoning & puzzles (seating, blood relations,
syllogisms), (3) Verbal ability (RC, grammar, para-jumbles), (4) Technical section for T&T;/Data roles (SQL, basic statistics, programming fundamentals), (5) Coding problems (1-2 medium/easy). Exact sections and counts
vary by intake (NLA / campus drives), but practice tests and previous patterns indicate a mix of ~60–90 minutes with adaptive sections and no negative marking in many cases. For up-to-date sample questions and placement-
paper style collections, popular resources include PrepInsta (Deloitte papers), DataLemur (SQL examples), InterviewBit (SQL cheat-sheets) and Deloitte's own practice pages for case interviews (links in appendix).

Quick study plan (6 weeks):

Week 1-2: Speed math, basics of probability, ratio, averages, DI practice. Week 3: Logical puzzles, seating arrangements, syllogisms. Week 4: Verbal — RC practice, grammar, para-jumbles. Week 5: SQL basics, window
functions, GROUP BY & joins. Week 6: Coding drills (two-sum, string problems), mock tests & revision.

2. Cheat-sheets

Quantitative Aptitude — Key formulas & shortcuts

• Percent change = (New - Old) / Old × 100
• Averages: Avg = Sum / n; Add an element x → new_avg = (old_sum + x) / (n+1)
• Speed/Distance/Time: speed = distance / time; for trains passing each other add lengths.
• Relative speed: same direction = v1 - v2; opposite = v1 + v2
• SI = (P × R × T) / 100; CI = P × (1 + R/100)^T - P
• Permutation nPr = n!/(n-r)! ; Combination nCr = n!/(r!(n-r)!)

Logical Reasoning — Common tips

• Draw diagrams for blood-relations and seating problems.
• Look for fixed anchors in circular arrangements (fix one person).
• Use Venn diagrams for syllogisms; check existential vs universal statements.
• For data sufficiency problems: evaluate statements individually and together without computing final numeric answer.

Verbal — Quick rules

• Tenses: 'did not go' (NOT 'did not went').
• Subject-verb agreement: Singular subjects → singular verbs (neither/each/every).
• Para-jumbles: identify opening sentence (introduces topic) and closing (summary/consequence).
• Reading Comprehension: skim questions first, then search for keywords in passage.

Technical — SQL & Python essentials

• SQL: GROUP BY, aggregates, subqueries and window functions are frequently tested.
• Common SQL tasks: second-highest salary, employees > avg salary, top-n per group, deduplication.
• Python: list/dict operations, two-pointer, hashing (dict for two-sum), string slicing.
• Statistics: mean, median, mode, variance, standard deviation basics.

Mock Test Set-2 (25 questions) — mixed sections

Q1. (Quant) A train 150m crosses a platform 250m in 40s. Find speed (km/hr).
Q2. (Quant) If A does job in 12 days, B in 20, with C taking 30 days; A and B work for 3 days then C joins. How many days total?
Q3. (Quant) Find x if 20% of x = 50.
Q4. (LR) Seating: In a row of 7, A sits 3rd from left, B 2nd from right. Who is in middle?
Q5. (LR) Syllogism: All A are B. Some B are C. Which follows?
Q6. (Verbal) RC: Short passage about remote work; Q: main idea?
Q7. (Verbal) Error spotting: 'Each of the players have a kit.'
Q8. (Verbal) Fill blank: She is good ___ analytics.
Q9. (Technical) SQL: Query employees with salary > AVG(salary).
Q10. (Technical) SQL: Second highest salary?
Q11. (Technical) Python: Reverse a string 'analytics' in one line.
Q12. (Technical) Python: Two-sum indices for [3,2,4], target=6.
Q13. (Quant) Probability: Draw 2 cards from deck. P(both hearts)?
Q14. (Quant) Boat speed downstream 14, upstream 10. Still water?
Q15. (Quant) Average: Average of 5 numbers is 20. Total?
Q16. (LR) Logical puzzle: Who sits where? (short)
Q17. (Technical) Coding: Find first non-repeating character in 'swiss'.
Q18. (Quant) Data Interpretation: Table percent change Q
Q19. (Verbal) Grammar: Choose correct sentence among options
Q20. (Verbal) Para-jumble: Arrange 4 lines
Q21. (Technical) Window function: Rank employees by salary per dept (SQL)
Q22. (Technical) DSA: Find duplicate in array of size n with numbers 1..n-1
Q23. (Quant) Probability: Two dice rolled. P(sum=7)
Q24. (LR) Misc reasoning: Constraint puzzle

Answers:

A1. 18 km/hr
A2. ~9.6 days (compute exact)
A3. 250
A4. 4th position (middle)
A5. Only 'Some B are C' given; 'Some A are C' not guaranteed
A6. Remote work increases flexibility
A7. 'have' → 'has'
A8. at
A9. SELECT * FROM emp WHERE salary > (SELECT AVG(salary) FROM emp);
A10. SELECT MAX(sal) FROM emp WHERE sal < (SELECT MAX(sal) FROM emp);
A11. 'analytics'[::-1]
A12. (1,2)
A13. (13/52)*(12/51)=1/17
A14. 12 km/hr
A15. 100
A16. (diagram-based answer)
A17. 'w'
A18. (answer in workbook)
A19. Option X
A20. Order provided
A21. Use ROW_NUMBER() OVER (PARTITION BY dept ORDER BY salary DESC)
A22. Use set or XOR
A23. 1/6

A24. Solution sketch

Mock Test Set-3 (25 questions) — mixed sections

Q1. (Quant) Find LCM of 12 and 18.
Q2. (Quant) A ratio 4:5 incomes; expenses 3:4; savings 200 and 300 respectively. Find incomes.
Q3. (Quant) Series: 3, 8, 15, 24, ?
Q4. (LR) Seating: Circular arrangement of 6 with alternating male/female - who opposite whom?
Q5. (LR) Syllogism: Some X are Y. All Y are Z. Conclusion?
Q6. (Verbal) RC: passage about data privacy; Q: key risk?
Q7. (Verbal) Error spotting: 'He and I went to the market.'
Q8. (Verbal) Fill blank: The manager, along with his team, ___ attending.
Q9. (Technical) SQL: Delete duplicate rows keeping one copy.
Q10. (Technical) SQL: Show top 3 salaries per department.
Q11. (Technical) Python: Check palindrome number 121.
Q12. (Technical) Python: Merge two sorted lists.
Q13. (Quant) Statistics: Find median of [1,3,3,6,7,8,9]
Q14. (Quant) Pipe leak fills tank in 10 hrs, empties in 15 hrs. Net?
Q15. (Quant) Average change problem
Q16. (LR) Logical puzzle: Arrange 5 houses by colour and pet
Q17. (Technical) Coding: Rotate array [1,2,3,4,5] by 2
Q18. (Quant) DI: Bar graph questions (percentages)
Q19. (Verbal) Grammar: 'whom' usage question
Q20. (Verbal) Para-jumble 5-sentence set
Q21. (Technical) SQL: Find employees who do not have managers (NULL manager_id)
Q22. (Technical) Hashing: Count pairs with diff k in array
Q23. (LR) Complex LR puzzle
Q24. (Technical) Coding: First missing positive in [3,4,-1,1]

Answers:

A1. 36
A2. Solve simultaneous equations
A3. 35
A4. (diagram-based)
A5. Some X are Z (valid)
A6. Unauthorized access
A7. Correct
A8. is
A9. Use CTE with ROW_NUMBER and delete rn>1
A10. Use ROW_NUMBER() OVER (PARTITION BY dept ORDER BY salary DESC)
A11. str(121)==str(121)[::-1]
A12. Two-pointer merge implementation
A13. 6
A14. 30 hrs
A15. (calculations)
A16. (answer)
A17. [4,5,1,2,3]
A18. (answers provided)
A19. Explanation
A20. Order provided
A21. WHERE manager_id IS NULL
A22. Use hashmap
A23. Solution outline
A24. 1

Mock Test Set-4 (25 questions) — mixed sections

Q1. (Quant) Simple vs compound interest for 2 years - which bigger and why?
Q2. (Quant) Pipes: A fills in 8h, B in 12h. A works 4h then B finishes. Total?
Q3. (Quant) 25% of a number is 15. Number?
Q4. (LR) Seating: Linear constraints - who is 2nd from left?
Q5. (LR) Syllogism: No A are B. Some C are B. Conclusion?
Q6. (Verbal) RC: cloud computing main benefit?
Q7. (Verbal) Error spotting: 'Each of the birds were singing.'
Q8. (Verbal) Fill blank: He is responsible ___ the data team.
Q9. (Technical) SQL: cumulative sum of salary ordered by date
Q10. (Technical) SQL: employees with exactly two direct reports
Q11. (Technical) Python: Remove vowels from 'education'
Q12. (Technical) Python: Count frequency using Counter
Q13. (Quant) Statistics: Mode of [2,2,2,3,3,4]
Q14. (Quant) Probability: with/without replacement questions
Q15. (Quant) Average of age groups question
Q16. (LR) Logic: Month/day matching puzzle
Q17. (Technical) Coding: Valid Sudoku row check
Q18. (Quant) DI: Pie chart percent question
Q19. (Verbal) Grammar: 'interested in' correct preposition
Q20. (Verbal) Para-jumble: 4-line ordering
Q21. (Technical) SQL: create view and rename column (example)
Q22. (Technical) SQL: find gaps in sequence of dates
Q23. (Technical) Python: flatten nested lists
Q24. (LR) LR: binary relations puzzle
Q25. (Technical) Coding: add two large numbers given as strings

Answers:

A1. CI greater unless rate=0
A2. Compute fractions
A3. 60
A4. (diagram answer)
A5. No direct relation between A and C
A6. Scalability and elasticity
A7. Use 'was'
A8. for
A9. SUM(salary) OVER (ORDER BY date)
A10. GROUP BY manager_id HAVING COUNT(*)=2
A11. List comprehension to filter vowels
A12. collections.Counter(list)
A13. 2
A14. (explain)
A15. (compute)
A16. (solution)
A17. Use set equality to compare digits
A18. (answers)
A19. in
A20. Order provided
A21. CREATE VIEW v AS SELECT col AS new_col FROM table;
A22. Use LAG/LEAD
A23. itertools.chain or recursion

A24. (answer)
A25. simulate addition

5. Coding Practice (10 problems) — descriptions + Python solutions

- Reverse words in a string. Ex: 'the sky is blue' -> 'blue is sky the'

```
def reverse_words(s):
return ' '.join(s.split()[::-1])
```

- Two-sum (indices).

```
def two_sum(nums, target):
seen = {}
for i,n in enumerate(nums):
if target-n in seen:
return (seen[target-n], i)
seen[n]=i
```

- Second largest number in array.

```
def second_largest(a):
uniq=sorted(set(a))
return uniq[-2] if len(uniq)>1 else None
```

- First non-repeating character in string 'swiss'.

```
def first_nonrep(s):
from collections import Counter
c=Counter(s)
for ch in s:
if c[ch]==1:
return ch
return None
```

- Rotate array by k.

```
def rotate(a,k):
k%=len(a)
return a[-k:]+a[:-k]
```

- Merge two sorted lists.

```
def merge(a,b):
i=j=0;res=[]
while i if a[i] else: res.append(b[j]); j+=1
res.extend(a[i:]); res.extend(b[j:]); return res
```

- Valid parentheses check.

```
def valid_paren(s):
stack=[]
pairs={'(':')','[':']','{':'}'}
for ch in s:
if ch in pairs: stack.append(ch)
else:
if not stack or pairs[stack.pop()]!=ch:
return False
return not stack
```

- Count pairs with difference k (hashmap).

```
def count_pairs(a,k):
s=set(a); cnt=0
for x in s:
if x+k in s: cnt+=1
return cnt
```

- First missing positive.

```
def first_missing_positive(a):
s=set(a)
i=1
while True:
if i not in s: return i
i+=1
```

- Add two large numbers as strings.

```
def add_str(a,b):
i=len(a)-1; j=len(b)-1; carry=0; res=[]
while i>=0 or j>=0 or carry:
x=int(a[i]) if i>=0 else 0
```

```
y=int(b[j]) if j>=0 else 0
s=x+y+carry
res.append(str(s%10)); carry=s//10
i-=1; j-=1
return ''.join(res[::-1])
```

6. SQL Challenges (20 exercises) — tasks + sample queries

- Employees with salary greater than average
```
SELECT * FROM employees WHERE salary > (SELECT AVG(salary) FROM employees);
```
- Second highest salary
```
SELECT MAX(salary) FROM employees WHERE salary < (SELECT MAX(salary) FROM employees);
```
- Count employees per dept having > 5 employees
```
SELECT dept_id, COUNT(*) FROM employees GROUP BY dept_id HAVING COUNT(*)>5;
```
- Managers earning less than direct reports
```
SELECT e.name FROM employees e JOIN employees m ON e.id=m.manager_id WHERE e.salary <
m.salary;
```
- Top N per group (dept top 3 salaries)
```
SELECT * FROM (SELECT e.*, ROW_NUMBER() OVER (PARTITION BY dept_id ORDER BY salary DESC) rn
FROM employees e) t WHERE rn<=3;
```
- Delete duplicate rows keeping one
```
WITH cte AS (SELECT *, ROW_NUMBER() OVER (PARTITION BY col1,col2 ORDER BY id) rn FROM table)
DELETE FROM table WHERE id IN (SELECT id FROM cte WHERE rn>1);
```
- Find employees without manager (root)
```
SELECT * FROM employees WHERE manager_id IS NULL;
```
- Cumulative salary over time (running sum)
```
SELECT date, salary, SUM(salary) OVER (ORDER BY date) AS running_sum FROM salaries;
```
- Find gaps in date sequence
```
SELECT date, LAG(date) OVER (ORDER BY date) prev_date FROM t;
```
- Identify VIP users (>=20 tasks/week consistently)
```
-- Aggregate by week and filter users whose weekly counts >=20 for all weeks in the month.
```
- Find employees who joined in last 30 days
```
SELECT * FROM employees WHERE join_date >= CURRENT_DATE - INTERVAL '30 days';
```
- Find customers with no orders
```
SELECT c.* FROM customers c LEFT JOIN orders o ON c.id=o.customer_id WHERE o.id IS NULL;
```
- Find most common product sold
```
SELECT product_id, COUNT(*) cnt FROM orders GROUP BY product_id ORDER BY cnt DESC LIMIT 1;
```
- Find running average of sales per month
```
SELECT month, AVG(sales) OVER (ORDER BY month ROWS BETWEEN UNBOUNDED PRECEDING AND CURRENT
ROW) FROM sales;
```
- Rank employees by salary per dept
```
SELECT name, dept_id, RANK() OVER (PARTITION BY dept_id ORDER BY salary DESC) rnk FROM
employees;
```
- Find overlapping date ranges
```
-- Use self-join on table where start1 <= end2 AND start2 <= end1 to find overlaps.
```
- Convert rows to columns (pivot)
```
-- Use CASE WHEN with aggregation or DB-specific PIVOT function.
```
- Find consecutive active days for a user
```
-- Use difference between date and ROW_NUMBER() to group streaks.
```
- Remove rows with NULL in critical column
```
DELETE FROM table WHERE critical_col IS NULL;
```
- Find median salary
```
-- Use percentile_cont(0.5) OVER () or DB-specific method to compute median.
```

Appendix — Recommended resources & sample-source sites

• PrepInsta — Deloitte past papers & mocks (aptitude, reasoning, coding).
https://prepinsta.com/deloitte/
• DataLemur — Deloitte SQL interview examples & practice.
https://datalemur.com/blog/deloitte-sql-interview-questions
• InterviewBit — SQL interview cheatsheet & practice.
https://www.interviewbit.com/sql-interview-questions/
• Deloitte case interview practice page (official).
https://caseinterviewprep.deloitte.com/
• GraduatesFirst — Deloitte assessment test guide.
https://www.graduatesfirst.com/deloitte-aptitude-tests